Low-Complexity Geometry-Aware Hybrid Beamforming for Near-Field STAR-RIS Networks

Nam Van Dinh*, Nguyen Tien Hoa†, Ngo Hoang Tu‡, Vo Nguyen Quoc Bao‡, Hai Minh Truong§, Thuan Van Le§
*Phenikaa School of Computing, Phenikaa University
†School of Electrical and Electronic Engineering, Hanoi University of Science and Technology, Vietnam
‡Faculty of Information Technology, Van Lang School of Technology, Van Lang University, Ho Chi Minh City, Vietnam
§Faculty of Electrical and Electronic Engineering, Phenikaa School of Engineering, Phenikaa University

***Abstract*—Large simultaneously transmitting and reflecting reconfigurable intelligent surface (STAR-RIS)-assisted wireless systems are increasingly likely to operate in the radiative near-field, where conventional far-field beam design becomes inadequate. This paper investigates low-complexity hybrid beamforming for large STAR-RIS near-field networks, where a base station serves users on both sides of an energy-splitting STAR-RIS. To exploit near-field geometry without incurring the high complexity of exhaustive search or fully-digital optimization, we propose a geometry-aware hybrid beamforming framework that combines proportional-fairness and beam-diversity-preserving user subset selection, side-aware energy splitting, common near-field STAR-RIS profile construction, and hybrid analog-baseband precoding. The proposed design is physically consistent with the common transmission/reflection profile constraint of practical STAR-RIS operation and remains scalable for large surfaces. Simulation results show that the proposed scheme outperforms practical hybrid benchmarks, significantly reduces the runtime compared with codebook-search baselines, and approaches the performance of a stronger genie-aided fully-digital upper benchmark.**

***Index Terms*—STAR-RIS, near-field communications, hybrid beamforming, user scheduling, energy splitting.**

I. Introduction

Reconfigurable intelligent surfaces (RISs) have emerged as a promising technology for shaping wireless propagation environments with low hardware cost and power consumption [1], [2]. By properly controlling the phase responses of many passive elements, RIS-assisted systems can improve coverage, spectral efficiency, and interference management without relying on fully active relays. Among various RIS architectures, simultaneously transmitting and reflecting RISs (STAR-RISs) are particularly attractive because they provide full-space coverage and support users on both sides of the surface [3], [4]. As a result, STAR-RISs have been widely studied for multi-user multiple-input multiple-output systems under energy-splitting and related operation modes [3], [4].

Meanwhile, the combination of large apertures and high carrier frequencies pushes RIS-assisted communications into the radiative near-field regime, where spherical-wave propagation must be considered and conventional far-field beamforming becomes suboptimal [5], [6]. In this regime, beam design depends jointly on angle and distance, and the effective multiuser channel structure changes significantly compared with the far-field case. This issue is particularly relevant for large STAR-RIS deployments, where the surface size itself makes near-field effects unavoidable for users located at practical service distances. Recent work has shown that near-field beamforming can provide substantial gains for STAR-RIS networks by exploiting additional spatial degrees of freedom [7].

Despite this potential, beamforming design for large STAR-RIS near-field networks remains challenging. The common transmission/reflection profiles of the STAR-RIS couple users on the same side of the surface, while near-field focusing enlarges the beam design space and complicates scheduling and profile construction [3], [7]. Although iterative alternating-optimization methods can achieve strong performance, their complexity quickly becomes prohibitive as the number of surface elements increases [7]. At the same time, fully-digital precoding, including weighted minimum-mean-square-error (MMSE)-based designs, provides a strong performance benchmark but is often impractical for large-array systems due to radio frequency (RF)-chain cost and power consumption [8]. This makes low-complexity hybrid beamforming (HBF) particularly appealing for large near-field deployments [9], [10].

Another important issue is user scheduling. Since only a limited number of streams can be supported in practical hybrid architectures, the scheduled subset largely determines both the multiplexing gain and the fairness behavior. A purely gain-based scheduler may repeatedly favor a few users with advantageous geometry, whereas a purely fairness-oriented rule may sacrifice too much beamforming efficiency. This tradeoff becomes even more critical in the near-field, where users with strong path gains may still produce highly correlated effective channels after common STAR-RIS profile construction [7]. Therefore, an effective low-complexity design should account not only for geometry-aware (GA) channel strength, but also for fairness and beam diversity during subset selection.

Motivated by these observations, this paper develops a low-complexity GA-HBF framework for large STAR-RIS near-field networks. The proposed design combines proportional-fairness and bitmask dynamic programming (PFBDP) user subset selection [11], side-aware energy splitting, common near-field STAR-RIS profile construction, and hybrid analog-

Corresponding author: Thuan Van Le (email: thuan.levan@phenikaa-uni.edu.vn).

baseband precoding. The main contributions are threefold:

1) **Near-field System Model:** We establish a physically consistent large STAR-RIS near-field multiuser downlink model with common transmission/reflection profiles and side-aware scheduling constraints, capturing the spherical-wave coupling that conventional far-field models overlook.
2) **Geometry-Aware Hybrid Beamforming:** We propose a GA-HBF scheme tailored to practical STAR-RIS operation, which integrates PFBDP-based user selection, side-aware energy splitting, and common near-field profile construction at low complexity.
3) **Performance Evaluation:** Extensive simulations show that the proposed design outperforms practical hybrid baselines, achieves a favorable complexity-performance tradeoff relative to near-field codebook search, and remains close to a stronger genie-aided fully-digital benchmark.

II. System Model

We consider a downlink large STAR-RIS-assisted near-field multiuser communication system, where a base station (BS) equipped with N_t antennas serves multiple single-antenna users with the aid of a STAR-RIS composed of $N = N_x N_y$ passive elements. The BS employs a uniform linear array (ULA), while the STAR-RIS is modeled as a uniform planar array (UPA) deployed on the yz-plane. The STAR-RIS operates in the energy-splitting mode, such that each element simultaneously supports transmission and reflection toward users located on the two sides of the surface. The setting is particularly suitable for full-space coverage, but it also introduces coupling among users sharing the same STAR-RIS side.

Let $\mathbf{p}_{\mathrm{B},m} \in \mathbb{R}^3$ and $\mathbf{p}_{\mathrm{R},n} \in \mathbb{R}^3$ denote the positions of the mth BS antenna and the nth STAR-RIS element, respectively, and let $\mathbf{p}_k \in \mathbb{R}^3$ denote the position of user k. The users are divided into the transmission-side set $\mathcal{K}_\mathrm{T}$ and the reflection-side set $\mathcal{K}_\mathrm{R}$, with $\mathcal{K}_\mathrm{T} \cup \mathcal{K}_\mathrm{R} = \mathcal{K}$ and $\mathcal{K}_\mathrm{T} \cap \mathcal{K}_\mathrm{R} = \emptyset$. To distinguish the two groups, we define the side indicator of user k as

$$s_k = \begin{cases} +1, & k \in \mathcal{K}_\mathrm{T}, \\ -1, & k \in \mathcal{K}_\mathrm{R}. \end{cases} \tag{1}$$

Since the considered STAR-RIS is electrically large and the users are assumed to lie in the radiative near-field region, both the BS–RIS and RIS–user links are described by spherical-wave propagation rather than the conventional planar-wave approximation.

Specifically, let $d_{\mathrm{B},m\to n} = \|\mathbf{p}_{\mathrm{B},m} - \mathbf{p}_{\mathrm{R},n}\|_2$ denote the distance from the mth BS antenna to the nth RIS element, where $\|\cdot\|_2$ is the Euclidean norm operator. The BS–RIS channel matrix $\mathbf{G} \in \mathbb{C}^{N\times N_t}$ is modeled as

$$[\mathbf{G}]_{n,m} = \frac{1}{\sqrt{N_t}} d_{\mathrm{B},m\to n}^{-\frac{\eta_{\mathrm{BR}}}{2}} \exp\left(-j\frac{2\pi}{\lambda} d_{\mathrm{B},m\to n}\right), \tag{2}$$

where λ is the carrier wavelength and η_{BR} is the BS–RIS path-loss exponent. Similarly, letting $d_{n\to k} = \|\mathbf{p}_{\mathrm{R},n} - \mathbf{p}_k\|_2$ denote the distance from the nth RIS element to user k, the RIS–user channel vector $\mathbf{r}_k \in \mathbb{C}^{N\times 1}$ is given by

$$[\mathbf{r}_k]_n = d_{n\to k}^{-\frac{\eta_{\mathrm{RU}}}{2}} \exp\left(-j\frac{2\pi}{\lambda} d_{n\to k}\right), \tag{3}$$

where η_{RU} denotes the RIS–user path-loss exponent. These expressions explicitly capture the distance-dependent phase and attenuation variations across the array aperture, which are essential in large near-field STAR-RIS systems.

To simultaneously serve users on both sides of the surface, the STAR-RIS forms one common transmission profile and one common reflection profile, denoted by $\mathbf{v}_\mathrm{T} = \sqrt{\beta_\mathrm{T}}\left[e^{j\phi_{\mathrm{T},1}}, \ldots, e^{j\phi_{\mathrm{T},N}}\right]^T$ and $\mathbf{v}_\mathrm{R} = \sqrt{\beta_\mathrm{R}}\left[e^{j\phi_{\mathrm{R},1}}, \ldots, e^{j\phi_{\mathrm{R},N}}\right]^T$, where $\beta_\mathrm{T}, \beta_\mathrm{R} \in [0,1]$ are the transmission and reflection energy-splitting factors satisfying $\beta_\mathrm{T} + \beta_\mathrm{R} = 1$. For user k, the effective STAR-RIS coefficient vector is chosen according to its side, i.e.,

$$\mathbf{v}_k = \begin{cases} \mathbf{v}_\mathrm{T}, & s_k = +1, \\ \mathbf{v}_\mathrm{R}, & s_k = -1. \end{cases} \tag{4}$$

Hence, the cascaded BS–RIS–user channel of user k can be written as

$$\mathbf{h}_k^H = \mathbf{r}_k^T \operatorname{diag}(\mathbf{v}_k)\mathbf{G} \in \mathbb{C}^{1\times N_t}. \tag{5}$$

This common-profile structure is a key feature of the considered model, since users on the same side are inherently coupled through the same transmission or reflection profile.

At the BS, hybrid precoding is adopted with N_{RF} RF chains, where $N_s \le N_{\mathrm{RF}} \le N_t$, and at most N_s users are served in each scheduling interval. Let $\mathcal{S} \subseteq \mathcal{K}$ denote the scheduled-user set with $|\mathcal{S}| = N_s' \le N_s$. The hybrid precoder is expressed as

$$\mathbf{F} = \mathbf{F}_{\mathrm{RF}}\mathbf{F}_{\mathrm{BB}}, \tag{6}$$

where $\mathbf{F}_{\mathrm{RF}} \in \mathbb{C}^{N_t\times N_{\mathrm{RF}}}$ and $\mathbf{F}_{\mathrm{BB}} \in \mathbb{C}^{N_{\mathrm{RF}}\times N_s'}$ denote the analog and digital precoders, respectively. The overall transmit power is constrained by $\|\mathbf{F}\|_F^2 \le P$, where $\|\cdot\|_F$ denotes the Frobenius norm operator. Let $\mathbf{s} \in \mathbb{C}^{N_s'\times 1}$ be the transmitted symbol vector with $\mathbb{E}[\mathbf{s}\mathbf{s}^H] = \mathbf{I}$. Then, the received signal at user $k \in \mathcal{S}$ is given by

$$y_k = \mathbf{h}_k^H \mathbf{F}\mathbf{s} + n_k, \tag{7}$$

where $n_k \sim \mathcal{CN}(0,\sigma^2)$ denotes additive white Gaussian noise. If $\mathbf{f}_k$ is the precoding vector associated with user k, the corresponding signal-to-interference-plus-noise ratio (SINR) is expressed as

$$\gamma_k = \frac{|\mathbf{h}_k^H\mathbf{f}_k|^2}{\sum_{\ell\in\mathcal{S},\,\ell\ne k}|\mathbf{h}_k^H\mathbf{f}_\ell|^2 + \sigma^2}, \tag{8}$$

which yields the achievable rate $R_k = \log_2(1+\gamma_k)$. Accordingly, the system sum-rate is $R_{\mathrm{sum}} = \sum_{k\in\mathcal{S}} R_k$.

The above model highlights the three main factors shaping the considered design problem: the near-field cascaded channel, the common transmission/reflection profiles imposed by the STAR-RIS, and the limited-stream HBF architecture at the BS.

III. Problem Formulation

Based on the system model, the considered design jointly determines the scheduled-user set, the common STAR-RIS transmission and reflection profiles, and the BS hybrid precoder so as to maximize the downlink spectral efficiency under practical hardware and scheduling constraints. Let $\mathcal{S} \subseteq \mathcal{K}$ denote the scheduled-user set. Since the BS supports only a limited number of streams, the number of simultaneously served users must satisfy $|\mathcal{S}| \leq N_s$. Moreover, to avoid an overly unbalanced allocation between the two sides of the STAR-RIS, per-side user caps are imposed as

$$|\mathcal{S} \cap \mathcal{K}_{\mathrm{T}}| \leq N_{\mathrm{T}}^{\max}, \ |\mathcal{S} \cap \mathcal{K}_{\mathrm{R}}| \leq N_{\mathrm{R}}^{\max}. \tag{9}$$

For the STAR-RIS, only one common transmission profile $\mathbf{v}_{\mathrm{T}}$ and one common reflection profile $\mathbf{v}_{\mathrm{R}}$ are applied in each scheduling interval, with

$$\beta_{\mathrm{T}} + \beta_{\mathrm{R}} = 1, \ 0 \leq \beta_{\mathrm{T}}, \beta_{\mathrm{R}} \leq 1, \tag{10}$$

while the BS hybrid precoder $\mathbf{F} = \mathbf{F}_{\mathrm{RF}}\mathbf{F}_{\mathrm{BB}}$ must satisfy

$$\|\mathbf{F}_{\mathrm{RF}}\mathbf{F}_{\mathrm{BB}}\|_F^2 \leq P. \tag{11}$$

Under these constraints, the design problem can be formulated as

$$\begin{aligned}
\max_{\mathcal{S}, \mathbf{v}_{\mathrm{T}}, \mathbf{v}_{\mathrm{R}}, \mathbf{F}_{\mathrm{RF}}, \mathbf{F}_{\mathrm{BB}}} \quad & \sum_{k \in \mathcal{S}} \log_2(1+\gamma_k) && (12)\\
\text{s.t.} \quad & |\mathcal{S}| \leq N_s, && (13)\\
& |\mathcal{S} \cap \mathcal{K}_{\mathrm{T}}| \leq N_{\mathrm{T}}^{\max}, && (14)\\
& |\mathcal{S} \cap \mathcal{K}_{\mathrm{R}}| \leq N_{\mathrm{R}}^{\max}, && (15)\\
& \beta_{\mathrm{T}} + \beta_{\mathrm{R}} = 1, \ \ 0 \leq \beta_{\mathrm{T}}, \beta_{\mathrm{R}} \leq 1, && (16)\\
& \|\mathbf{F}_{\mathrm{RF}}\mathbf{F}_{\mathrm{BB}}\|_F^2 \leq P. && (17)
\end{aligned}$$

This problem is challenging because the scheduled-user set is combinatorial, the effective channels are coupled through the common STAR-RIS profiles, and the hybrid precoder introduces additional nonconvexity through the analog-baseband factorization.

Although (12) focuses on throughput maximization, purely gain-driven scheduling may repeatedly favor a small set of geometrically advantageous users, especially when the stream budget is limited. To alleviate this issue, fairness is incorporated at the scheduling stage rather than replacing the sum-rate objective itself. Let T_k denote the long-term average served rate of user k. A proportional-fairness (PF) metric is defined as

$$\psi_k^{\mathrm{PF}} = \frac{\widehat{R}_k}{T_k + \epsilon}, \tag{18}$$

where $\widehat{R}_k$ is a geometry-based utility estimate and $\epsilon > 0$ is a small regularization constant. In the considered near-field setting, $\widehat{R}_k$ is obtained from the large-scale cascaded geometry and the effective focusing potential of user k, rather than from costly iterative beam optimization.

However, maximizing (18) alone may still produce highly correlated effective channels and hence poor multiuser spatial separation. Therefore, the scheduling stage should favor user subsets that are not only geometrically strong and fairness-aware, but also sufficiently decorrelated in the effective channel domain. This motivates the proposed PF-based BDP user assignment, where proportional-fairness utilities are combined with channel-correlation penalties before the common STAR-RIS profiles and hybrid precoder are constructed.

IV. Proposed Geometry-Aware Hybrid Beamforming

The proposed design follows the problem structure in Section III and consists of four stages: fairness- and diversity-aware user subset selection, side-aware energy splitting, common near-field STAR-RIS profile construction, and hybrid analog-baseband precoding.

The first stage determines the scheduled-user subset under the stream-budget and side-cap constraints. For each user $k \in \mathcal{K}$, we define a GA utility proxy $\widehat{G}_k$ from the cascaded BS–RIS and RIS–user propagation, and construct the proportional-fairness score

$$\psi_k = \frac{\widehat{G}_k}{T_k + \epsilon}, \tag{19}$$

where T_k is the long-term average served rate and $\epsilon > 0$ is a small regularization constant. To avoid selecting users with highly correlated effective channels, we further introduce the correlation measure

$$\rho_{i,j} = \frac{|\widetilde{\mathbf{h}}_i \widetilde{\mathbf{h}}_j^H|}{\|\widetilde{\mathbf{h}}_i\|_2 \|\widetilde{\mathbf{h}}_j\|_2 + \varepsilon_c}, \tag{20}$$

where $\widetilde{\mathbf{h}}_k$ is an approximate effective channel proxy and $\varepsilon_c > 0$ avoids numerical instability.

Using (19) and (20), the subset utility is defined as

$$\Psi(\mathcal{S}) = \sum_{k \in \mathcal{S}} \psi_k - \mu \sum_{i,j \in \mathcal{S}, i<j} \rho_{i,j}, \tag{21}$$

where $\mu \geq 0$ controls the tradeoff between fairness and beam diversity. The scheduled-user set is then obtained by maximizing (21) over a pruned candidate pool subject to the stream-budget and side-cap constraints. Since N_s is small in practical hybrid architectures, this search remains manageable.

After obtaining the scheduled-user set $\mathcal{S}$, the transmission-side and reflection-side energy-splitting factors are determined from the aggregate utility demands on the two sides:

$$D_{\mathrm{T}} = \sum_{k \in \mathcal{S} \cap \mathcal{K}_{\mathrm{T}}} \widehat{G}_k, \ D_{\mathrm{R}} = \sum_{k \in \mathcal{S} \cap \mathcal{K}_{\mathrm{R}}} \widehat{G}_k. \tag{22}$$

The raw splitting factors are chosen as

$$\beta_{\mathrm{T}} = \frac{D_{\mathrm{T}}}{D_{\mathrm{T}} + D_{\mathrm{R}} + \varepsilon_\beta}, \ \beta_{\mathrm{R}} = 1 - \beta_{\mathrm{T}}, \tag{23}$$

and then clipped to satisfy a minimum energy floor.

Given $\mathcal{S}$ and $(\beta_{\mathrm{T}}, \beta_{\mathrm{R}})$, the STAR-RIS transmission and reflection profiles are constructed from the aggregate near-field geometry of the selected users. Specifically,

$$\mathbf{a}_{\mathrm{T}} = \sum_{k\in\mathcal{S}\cap\mathcal{K}_{\mathrm{T}}} w_k \frac{\mathbf{r}_k}{|\mathbf{r}_k| + \varepsilon_r},$$
$$\mathbf{a}_{\mathrm{R}} = \sum_{k\in\mathcal{S}\cap\mathcal{K}_{\mathrm{R}}} w_k \frac{\mathbf{r}_k}{|\mathbf{r}_k| + \varepsilon_r}, \tag{24}$$

where the division is element-wise and w_k is proportional to the user utility. The common STAR-RIS profiles are then set as

$$\mathbf{v}_{\mathrm{T}} = \sqrt{\beta_{\mathrm{T}}}\exp\big(-j\angle(\mathbf{a}_{\mathrm{T}})\big),$$
$$\mathbf{v}_{\mathrm{R}} = \sqrt{\beta_{\mathrm{R}}}\exp\big(-j\angle(\mathbf{a}_{\mathrm{R}})\big), \tag{25}$$

which preserves the common-profile STAR-RIS structure while exploiting spherical-wave near-field focusing.

With the STAR-RIS profiles fixed, the effective channel of each scheduled user is

$$\mathbf{h}_k^H = \mathbf{r}_k^T \operatorname{diag}(\mathbf{v}_k)\mathbf{G}, \tag{26}$$

where $\mathbf{v}_k = \mathbf{v}_{\mathrm{T}}$ for $k \in \mathcal{K}_{\mathrm{T}}$ and $\mathbf{v}_k = \mathbf{v}_{\mathrm{R}}$ for $k \in \mathcal{K}_{\mathrm{R}}$. Stacking the scheduled-user channels gives

$$\mathbf{H}_{\mathcal{S}} = \left[\mathbf{h}_{k_1}^H, \mathbf{h}_{k_2}^H, \dots, \mathbf{h}_{k_{N_s'}}^H\right]^T. \tag{27}$$

The analog RF precoder is obtained by phase-only alignment with the scheduled-user effective channels:

$$[\mathbf{F}_{\mathrm{RF}}]_{:,i} = \frac{1}{\sqrt{N_t}}\exp\big(j\angle(\mathbf{h}_{k_i}^*)\big), \quad i = 1,\dots,N_s'. \tag{28}$$

Then, the reduced-dimensional baseband channel is $\mathbf{H}_{\mathrm{BB}} = \mathbf{H}_{\mathcal{S}}\mathbf{F}_{\mathrm{RF}}$, and the digital precoder is designed using regularized zero-forcing (RZF) such that

$$\mathbf{F}_{\mathrm{BB}} = \mathbf{H}_{\mathrm{BB}}^H\big(\mathbf{H}_{\mathrm{BB}}\mathbf{H}_{\mathrm{BB}}^H + \lambda_{\mathrm{RZF}}\mathbf{I}\big)^{-1}. \tag{29}$$

The final hybrid precoder is

$$\mathbf{F} = \mathbf{F}_{\mathrm{RF}}\mathbf{F}_{\mathrm{BB}}, \ \mathbf{F} \leftarrow \sqrt{P}\frac{\mathbf{F}}{\|\mathbf{F}\|_F}. \tag{30}$$

This design combines low-complexity phase-only RF steering with interference-aware digital precoding.

The proposed method avoids full alternating optimization over the user subset, STAR-RIS coefficients, and hybrid precoder. Its dominant complexity comes from GA utility computation, PF-BDP subset search over a pruned candidate pool, common profile construction, and low-dimensional RZF precoding. Accordingly, the overall per-slot complexity can be summarized as

$$\mathcal{O}\left(KN + \sum_{m=1}^{N_s}\binom{L_c}{m}m^2 + N_sN + (N_s')^3\right), \tag{31}$$

where L_c denotes the candidate-pool size. Since N_s and N_s' are small in practical hybrid systems, the proposed framework remains substantially more scalable than exhaustive codebook-search and fully-digital benchmark schemes.

Algorithm 1 summarizes the overall GA-HBF-PFBDP procedure as follows. The method first identifies a fairness-aware and diversity-preserving user subset, then determines the side-aware STAR-RIS energy split and the corresponding common near-field transmission/reflection profiles, and finally constructs the hybrid analog-baseband precoder from the resulting effective channels. Since the design avoids full alternating optimization over all RIS coefficients and beamforming variables, it remains computationally scalable for large STAR-RIS configurations while preserving most of the attainable near-field beamforming gain. In the next section, simulation results are provided to evaluate the spectral-efficiency, fairness, and runtime performance of the proposed scheme against several practical and genie-aided benchmark designs.

Algorithm 1: Proposed GA-HBF-PFBDP Scheme

Input: BS–RIS channel $\mathbf{G}$, RIS–user channels $\{\mathbf{r}_k\}$, long-term rates $\{T_k\}$, stream budget N_s, side caps $N_{\mathrm{T}}^{\max}, N_{\mathrm{R}}^{\max}$, transmit power P

Output: Scheduled-user set $\mathcal{S}$, STAR-RIS profiles $\mathbf{v}_{\mathrm{T}}, \mathbf{v}_{\mathrm{R}}$, hybrid precoder $\mathbf{F}$

Compute $\widehat{G}_k$, ψ_k in (19), and $\rho_{i,j}$ in (20);
Select $\mathcal{S}$ by maximizing (21) under (13)–(15);
Compute $(\beta_{\mathrm{T}}, \beta_{\mathrm{R}})$ from (22) and (23);
Construct $\mathbf{v}_{\mathrm{T}}$ and $\mathbf{v}_{\mathrm{R}}$ using (24) and (25);
Build $\{\mathbf{h}_k\}_{k\in\mathcal{S}}$ from (26);
Construct $\mathbf{F}_{\mathrm{RF}}$ from (28), compute $\mathbf{F}_{\mathrm{BB}}$ via (29), and normalize $\mathbf{F}$ using (30);
Return $\mathcal{S}$, $\mathbf{v}_{\mathrm{T}}$, $\mathbf{v}_{\mathrm{R}}$, and $\mathbf{F}$;

V. Simulation Results and Discussions

In this section, simulation results are presented to evaluate the spectral-efficiency and computational-efficiency performance of the proposed GA-HBF-PFBDP scheme in large STAR-RIS near-field networks. Unless otherwise stated, all curves are averaged over independent random user drops. Unless otherwise stated, the main simulation parameters are summarized in Table I.

To assess the effectiveness of the proposed design, we compare it with four representative baselines under the same physical model. NF-HBF-TopGain-MMSE uses the same near-field cascaded channel and common STAR-RIS profile construction, but replaces PF-BDP scheduling by simple top-gain user selection. NF-Codebook-Greedy constructs the STAR-RIS transmission and reflection profiles through near-field focal-point codebook search followed by greedy scheduling and hybrid precoding. FF-HBF-MMSE adopts conventional far-field phase-profile construction with HBF. Finally, FD-NF-SVD-UB serves as a genie-aided fully-digital upper benchmark to indicate how closely the proposed low-complexity design approaches a stronger fully-digital solution. Table II summarizes the characteristics of the considered benchmark schemes.

Fig. 1 shows the average sum-rate versus transmit signal-to-noise ratio (SNR). The proposed GA-HBF-PFBDP consistently achieves the best performance among the practical hybrid

Table I
MAIN SIMULATION PARAMETERS

Parameter	**Value**
Carrier frequency f_c	28 GHz
Wavelength λ	c/f_c
BS array structure	ULA
Number of BS antennas N_t	16
Default STAR-RIS size	$N_x = N_y = 16$
BS and RIS spacing	$\lambda/2$
BS position	$(0, 0, 0)$ m
STAR-RIS center	$(8, 0, 0)$ m
Default users per side	4 T, 4 R
Maximum scheduled streams N_s	4
Per-side scheduling cap	2 users/side
Noise variance σ^2	1
BS–RIS path-loss exponent η_{BR}	2.0
RIS–user path-loss exponent η_{RU}	2.2
Energy-splitting floor $\beta_{\min}$	0.15

Table II
BENCHMARK CHARACTERISTICS

Schemes	NF	HBF	Scheduling
Proposed GA-HBF-PFBDP	Yes	Yes	PF-BDP
NF-HBF-TopGain-MMSE	Yes	Yes	Top-gain
NF-Codebook-Greedy	Yes	Yes	Greedy
FF-HBF-MMSE	No	Yes	Gain-based
FD-NF-SVD-UB	Yes	No	Genie-aided

schemes and remains close to the genie-aided FD-NF-SVD-UB benchmark across the entire SNR range. This indicates that the proposed GA-HBF design captures most of the near-field beamforming gain with substantially lower complexity than a fully-digital solution. Compared with NF-HBF-TopGain-MMSE, the proposed method achieves a clear improvement, especially at moderate and high SNR, which confirms the benefit of PF-BDP scheduling over simple top-gain selection. NF-Codebook-Greedy remains competitive but inferior to the proposed design, while FF-HBF-MMSE performs the worst, showing that conventional far-field beamforming is inadequate in large STAR-RIS near-field scenarios.

Fig. 2 plots the average sum-rate versus the number of STAR-RIS elements. As the RIS size increases, all near-field schemes benefit from the larger effective aperture and stronger spatial focusing capability. The proposed GA-HBF-PFBDP scheme consistently tracks the fully-digital upper benchmark and outperforms the practical baselines across all RIS sizes, showing that it efficiently exploits the additional near-field degrees of freedom offered by large STAR-RIS arrays. In contrast, NF-Codebook-Greedy grows more slowly because its predefined focal-point candidates cannot fully adapt to the continuous near-field geometry, while NF-HBF-TopGain-MMSE quickly saturates due to its simple gain-based scheduling. FF-HBF-MMSE again exhibits the weakest scaling behavior, further confirming the mismatch of far-field beam design in this regime.

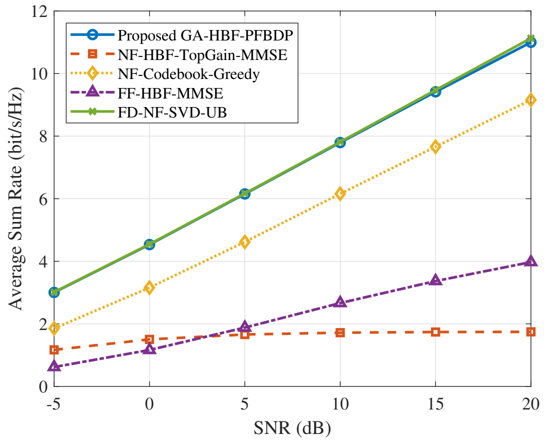

Figure 1. Average sum-rate versus SNR for different beamforming schemes in a large STAR-RIS near-field system.

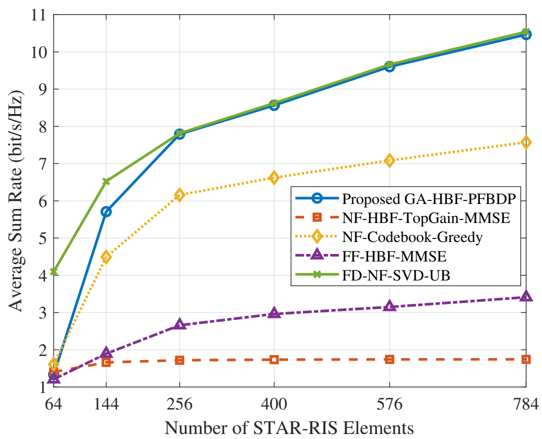

Figure 2. Average sum-rate versus the number of STAR-RIS elements for different beamforming schemes.

The strong performance of the proposed GA-HBF-PFBDP scheme can be explained by the structure of the considered near-field STAR-RIS system. First, the common transmission and reflection profiles largely determine the cascaded channel quality, so a GA profile design already captures a dominant portion of the available focusing gain. Second, because only a small number of streams is simultaneously served, the loss introduced by hybrid precoding remains limited once the scheduled-user subset is sufficiently decorrelated. Third, the PF-BDP scheduling stage improves not only fairness awareness but also the effective spatial compatibility among the served users, which directly benefits the subsequent analog-baseband precoder design. As a result, the proposed framework is able to approach a much stronger fully-digital benchmark while maintaining substantially lower complexity.

Fig. 3 compares the average runtime per drop as the RIS size increases. The proposed scheme maintains a nearly flat runtime profile, which confirms its computational scalability for large STAR-RIS deployments. This is mainly because it avoids exhaustive focal-point search and full alternating optimization, and instead relies on GA profile construction

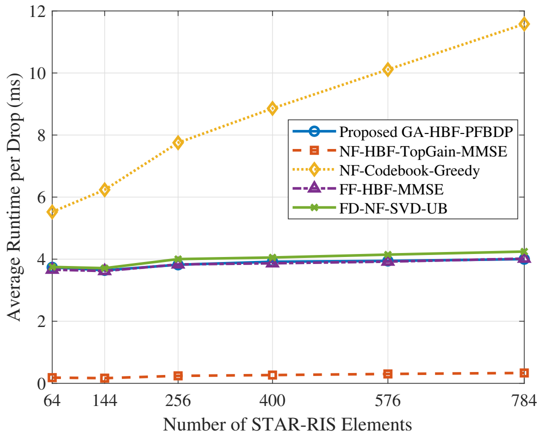

Figure 3. Average runtime per drop versus the number of STAR-RIS elements for different beamforming schemes.

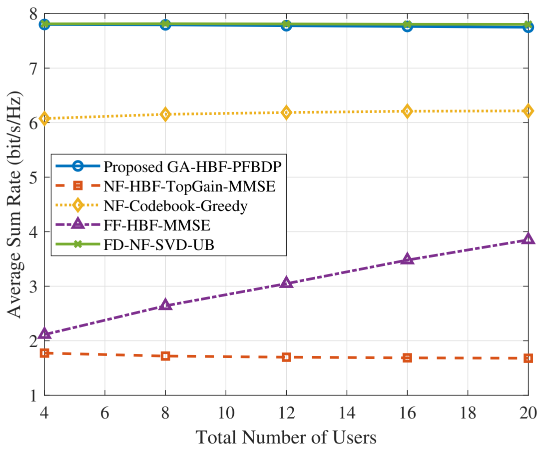

Figure 4. Average sum-rate versus the total number of users for different beamforming schemes.

and low-dimensional hybrid precoding. NF-Codebook-Greedy has the highest runtime and grows the fastest with the surface size due to repeated codebook search. Although NF-HBF-TopGain-MMSE has the lowest complexity, this comes at the cost of substantial performance loss. Overall, the proposed method achieves a favorable runtime-performance tradeoff, combining near-upper-bound sum-rate performance with significantly lower complexity than codebook-search and fully-digital benchmarks.

Fig. 4 shows the average sum-rate versus the total number of users under a fixed stream budget. The proposed GA-HBF-PFBDP scheme consistently achieves the highest sum-rate among the practical hybrid baselines and remains close to the fully-digital benchmark over the entire user range. This indicates that the proposed fairness-aware and BDP subset selection can effectively exploit multiuser diversity while maintaining strong near-field focusing. NF-Codebook-Greedy achieves the second-best practical performance, but its gain saturates more quickly as the user pool grows because its fixed focal-point structure cannot fully adapt to the continuously varying user geometry. NF-HBF-TopGain-MMSE shows little improvement with more users, revealing that simple top-gain scheduling cannot convert a larger candidate pool into effective multiplexing gain. FF-HBF-MMSE improves with the number of users but remains significantly below the near-field schemes, again highlighting the importance of GA near-field beam design.

VI. Conclusion

This paper investigated low-complexity HBF for large STAR-RIS near-field networks. By jointly combining PF-BDP user subset selection, side-aware energy splitting, common near-field STAR-RIS profile construction, and hybrid analog-baseband precoding, the proposed GA-HBF-PFBDP framework was shown to effectively exploit the geometric structure of near-field propagation while remaining computationally scalable. Simulation results demonstrated that the proposed scheme consistently outperforms practical hybrid benchmarks, achieves a favorable runtime-performance tradeoff compared with codebook-search baselines, and closely approaches the performance of a stronger genie-aided fully-digital benchmark. These results confirm that most of the attainable near-field beamforming gain in large STAR-RIS systems can be harvested without resorting to high-complexity joint optimization.

References

[1] M. A. ElMossallamy, H. Zhang, L. Song, K. G. Seddik, Z. Han, and G. Y. Li, "Reconfigurable intelligent surfaces for wireless communications: Principles, challenges, and opportunities," *IEEE Trans. Cogn. Commun. Netw.*, vol. 6, no. 3, pp. 990–1002, 2020.

[2] X. Yuan, Y.-J. A. Zhang, Y. Shi, W. Yan, and H. Liu, "Reconfigurable-intelligent-surface empowered wireless communications: Challenges and opportunities," *IEEE Wireless Commun.*, vol. 28, no. 2, pp. 136–143, 2021.

[3] H. Niu, Z. Chu, F. Zhou, P. Xiao, and N. Al-Dhahir, "Simultaneous transmission and reflection reconfigurable intelligent surface assisted mimo systems," 2021. [Online]. Available: https://arxiv.org/abs/2106.09450

[4] Y. Liu, J. Xu, Z. Wang, X. Mu, J. Zhang, and P. Zhang, "Simultaneously transmitting and reflecting (STAR) RISs for 6G: Fundamentals, recent advances, and future directions," *Frontiers Inf. Technol. Electron. Eng.*, vol. 24, no. 12, pp. 1689–1707, 2023.

[5] S. Hu, H. Wang, and M. C. Ilter, "Design of near-field beamforming for large intelligent surfaces," *IEEE Trans. Wireless Commun.*, vol. 23, no. 1, pp. 762–774, 2024.

[6] Y. Liu, C. Ouyang, Z. Wang, J. Xu, X. Mu, and A. L. Swindlehurst, "Near-field communications: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 3, pp. 1687–1728, 2025.

[7] H. Li, Y. Liu, X. Mu, Y. Chen, Z. Pan, and Y. C. Eldar, "Near-field beamforming for STAR-RIS networks," 2023. [Online]. Available: https://arxiv.org/abs/2306.14587

[8] S. S. Christensen, R. Agarwal, E. de Carvalho, and J. M. Cioffi, "Weighted sum-rate maximization using weighted mmse for MIMO-BC beamforming design," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2009, pp. 1–6.

[9] C. Meng, D. Ma, Z. Wang, Y. Liu, Z. Wei, and Z. Feng, "Near-field hybrid beamforming design for modular XL-MIMO ISAC systems," *IEEE Trans. Commun.*, vol. 73, no. 11, pp. 11 840–11 854, 2025.

[10] M. Liu, M. Li, R. Liu, and Q. Liu, "Dynamic hybrid beamforming designs for ELAA near-field communications," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 3, pp. 644–658, 2025.

[11] T. V. Le, K. Lee, and N. Cong Luong, "Bitmask dynamic programming for user scheduling in multi-user MIMO mmWave systems," *IEEE Commun. Lett.*, vol. 27, no. 12, pp. 3365–3369, 2023.